

# A016 — Helmholtz Vorticity-Population / Matter–Field Partition Falsifier

## Falsification Protocol and Final Report

SST-Workbench

Version v0.2.1

### 1 Identification, scope, and status

| Field      | Value                                                                                                                               |
|------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Catalog ID | A016                                                                                                                                |
| Falsifier  | helmholtz_vorticity_population_matter_field_partition                                                                               |
| Version    | v0.2.1                                                                                                                              |
| Framework  | SST Falsifier Framework v1.0.6 / SST-FALSIFIER-RUNTIME-2                                                                            |
| Profile    | cpu: Python FP64 reference plus strict C++/OpenMP FP64 certification                                                                |
| Status     | Frozen protocol is created by <code>run_all.cmd FREEZE</code> ; later results are inserted automatically without editing this text. |

This version preserves the numerical thresholds of A016 v0.1.1/v0.2.0, replaces the single direct KnotPlot input route by the qualified E010/PKLSA v0.4.0 geometry layer, and adds a provider-level cross-source consistency gate. A physical/scientific gate failure does not terminate sibling scientific branches.

### 2 Research question and objective

Can a smooth incompressible inviscid vortex model simultaneously satisfy: (i) material preservation of the zero/nonzero vorticity population; (ii) material-volume and vorticity-flux conservation; (iii) a locally irrotational but nontrivial exterior velocity field with circulation periods and nonlocal kinetic energy; and (iv) the original static relaxed-centerline relative-equilibrium test?

The objective is not to prove an SST interpretation. It is to make the Helmholtz-to-SST partition testable with explicit observables, provenance, convergence, backend certification, and geometry-provider robustness. The static centerline branch cannot establish actual long-time Euler persistence or reconnection avoidance. PKLSA is used as a geometry qualification/provenance layer and is not treated as an Euler-stability proof.

### 3 Competing hypotheses and allowed conclusions

**Null hypothesis  $H_0$ .** At least one preregistered consequence fails beyond tolerance, or the H2/H3/P3/P4/P5/P6 classifications are not reproducible across the required distinct upstream PKLSA provider groups for the same topology. Thus the material controls, exterior-field branch, unchanged centerline branch, and cross-source robustness branch are independently falsifiable.

**Alternative hypothesis  $H_1$ .** The exact material controls close numerically; the exterior is locally divergence-free/curl-free but can retain nonzero speed and circulation periods; a resolved energy fraction lies outside the near-core source zone; eligible closed sources have stable far-field scaling; each qualified PKLSA carrier receives an H0–H4/P3–P6 classification; and, wherever at least two distinct upstream provider groups exist for one topology, X0 satisfies its preregistered cross-source agreement criterion.

**Falsification semantics.** Each H\* or P\* branch reports independently after G0/G1/G2. Aggregate FAIL means at least one eligible input failed that branch; per-input results remain in `BLIND_SAMPLE_METRICS.json`. X0 is a separate provider-consistency result and does not overwrite the per-carrier classifications. Backend failure is certification evidence only. No threshold may be moved after freeze.

## 4 Assumptions, domain of validity, and known edge cases

- A1. Smooth three-dimensional incompressible Euler flow, no viscosity, no external nonconservative forcing.
- A2. Source geometries are closed oriented filaments. Finite-core regularization is used only where self-interaction/energy would otherwise be singular.
- A3. The blind geometry branch uses PKLSA-qualified carriers in source units and unit circulation. No physical density or absolute length calibration participates in blind scoring.
- A4. PKLSA-generated/derived carriers may be measured but cannot close the X0 cross-source gate; multiple variants from one provider group collapse to one provider vote.
- A5. P3 samples only exterior points. P5 uses a geometric near-core mask and therefore tests nonlocal velocity-energy support, not exact compact vorticity support.
- A6. P6 excludes near-zero-vector-area geometries because cancellation may remove the leading localized-source far-field moment.

The experiment does not cover viscosity, compressibility, reconnection, singular Euler breakdown, or externally forced vorticity production.

## 5 Symbols, units, and dimensional consistency

| Symbol                | Meaning                              | SI unit                    | Dimension    |
|-----------------------|--------------------------------------|----------------------------|--------------|
| $\mathbf{x}$          | position                             | m                          | $L$          |
| $t$                   | flow time                            | s                          | $T$          |
| $\mathbf{u}$          | velocity                             | $\text{m s}^{-1}$          | $LT^{-1}$    |
| $\boldsymbol{\omega}$ | vorticity $\nabla \times \mathbf{u}$ | $\text{s}^{-1}$            | $T^{-1}$     |
| $\mathbf{F}$          | deformation gradient                 | 1                          | 1            |
| $J$                   | $\det \mathbf{F}$                    | 1                          | 1            |
| $\Gamma$              | circulation/vorticity flux           | $\text{m}^2 \text{s}^{-1}$ | $L^2 T^{-1}$ |
| $a$                   | core radius                          | m after scale choice       | $L$          |
| $\ell_E$              | $E/(\rho\Gamma^2)$                   | m                          | $L$          |
| $p$                   | velocity far-field exponent          | 1                          | 1            |

All machine-used equations and dimensional checks are duplicated in the frozen `science_contract.json` and rendered below.

## 6 Exact mathematical model and observables

The central continuum relations are

$$\boldsymbol{\omega} = \nabla \times \mathbf{u}, \quad (1)$$

$$\boldsymbol{\omega}(\mathbf{X}(\mathbf{a}, t), t) = \mathbf{F}(\mathbf{a}, t)\boldsymbol{\omega}_0(\mathbf{a}), \quad (2)$$

$$J = \det \mathbf{F} = 1, \quad (3)$$

$$\Phi_\omega = \int_{S(t)} \boldsymbol{\omega} \cdot d\mathbf{S} = \Gamma = \text{const.} \quad (4)$$

For the numerical filament reconstruction,

$$\mathbf{u}(\mathbf{x}) = \frac{\Gamma}{4\pi} \oint_C \frac{d\boldsymbol{\ell}' \times (\mathbf{x} - \mathbf{x}')}{(|\mathbf{x} - \mathbf{x}'|^2 + a^2)^{3/2}}, \quad (5)$$

with  $a = 0$  only for exterior evaluations that remain separated from the filament. The energy-length coefficient is

$$E = \rho\Gamma^2\ell_E, \quad \ell_E = \frac{1}{8\pi} \oint \oint \frac{d\boldsymbol{\ell} \cdot d\boldsymbol{\ell}'}{\sqrt{|\mathbf{x} - \mathbf{x}'|^2 + a^2}}. \quad (6)$$

Relative equilibrium is tested after fitting rigid translation, rigid rotation, and tangential reparameterization. Exterior topology is tested through

$$\Gamma^{-1} \oint_\gamma \mathbf{u} \cdot d\boldsymbol{\ell} \approx \text{Lk}(\gamma, C). \quad (7)$$

The exterior differential audit tests  $\nabla \cdot \mathbf{u} = 0$  and  $\nabla \times \mathbf{u} = 0$ . The energy partition observable is

$$f_{\text{ext}} = \frac{\int_{V \setminus V_{\text{core}}} \frac{1}{2} |\mathbf{u}|^2 dV}{\int_V \frac{1}{2} |\mathbf{u}|^2 dV}. \quad (8)$$

Far-field shells fit

$$\log u_{\text{rms}}(r) = b - p \log r. \quad (9)$$

A steady irrotational Bernoulli interpretation maps the frozen velocity exponent to  $|\Delta p| \propto r^{-2p}$  and  $|\nabla p| \propto r^{-(2p+1)}$ ; this is a post-fit interpretation, not a fitting constraint. For X0, a provider-collapsed classification agreement is

$$A_{T,g} = \frac{\max_s n_s(T, g)}{N_{\text{provider}}(T, g)}, \quad (10)$$

where  $s$  is one of PASS, FAIL, INELIGIBLE, or INTERNAL\_INCONSISTENCY. The frozen acceptance threshold is  $A_{T,g} \geq 0.80$  with at least two distinct upstream provider groups.

### Machine-readable science contract

**Research question** Within smooth incompressible inviscid vortex dynamics, can a fixed material population carrying nonzero vorticity coexist with an exterior locally irrotational sector that still carries nonzero velocity, circulation periods, kinetic energy, and a reproducible far field, while the original A016 relaxed-centerline relative-equilibrium test remains independently falsifiable?, and are those geometry-dependent classifications robust across distinct qualified PKLSA source/provider families for the same topology?

**Objective** Fuse the v0.1.1 Helmholtz centerline gates with theorem-level Cauchy-map controls and explicit matter-field-partition observables, without retuning the original H0-H4 thresholds and without allowing an early scientific FAIL to suppress later independent measurements. v0.2.1 additionally tests whether the matter-field conclusions are representation/provider dependent within the PKLSA v0.4.0 qualified atlas.

**H<sub>0</sub>** At least one registered ideal-flow consequence or centerline condition fails beyond its preregistered tolerance, or the resulting H2/H3/P3/P4/P5/P6 classifications are not reproducible across the required distinct upstream PKLSA provider groups for the same topology. Material zero/nonzero vorticity classification, material volume, vorticity flux, exterior potentiality, exterior energy support, far-field scaling, relaxed-centerline relative equilibrium, and cross-source provider consistency are therefore independently falsifiable.

**H<sub>1</sub>** The exact material controls satisfy Cauchy/volume/flux invariance; the reconstructed exterior is locally divergence-free and curl-free but nontrivial; a resolved nonzero kinetic-energy fraction lies outside the near-core source zone; eligible closed filaments show stable localized-source far-field scaling; each qualified PKLSA carrier receives an independent H0-H4/P3-P6 classification without threshold adjustment; and, for topologies with at least two distinct upstream provider groups, the preregistered X0 agreement criterion is satisfied.

## Registered equations

### E1: Define vorticity

$$\boldsymbol{\omega} = \nabla \times \mathbf{u} \quad (11)$$

*Dimensional check:* curl contributes  $L^{-1}$ , so  $[\boldsymbol{\omega}] = (L T^{-1}) L^{-1} = T^{-1}$ .

### E2: Cauchy vorticity map for incompressible Euler

$$\boldsymbol{\omega}(\mathbf{X}(\mathbf{a}, t), t) = \mathbf{F}(\mathbf{a}, t) \boldsymbol{\omega}_0(\mathbf{a}), \quad \mathbf{F} = \partial \mathbf{X} / \partial \mathbf{a} \quad (12)$$

*Dimensional check:*  $\mathbf{F}$  is dimensionless and both vorticities have  $T^{-1}$ .

### E3: Incompressible material-volume constraint

$$J = \det \mathbf{F} = 1 \quad (13)$$

*Dimensional check:* determinant of a dimensionless deformation gradient is dimensionless.

### E4: Vorticity-flux/circulation conservation control

$$\Phi_{\boldsymbol{\omega}} = \int_{S(t)} \boldsymbol{\omega} \cdot d\mathbf{S} = \Gamma = \text{const} \quad (14)$$

*Dimensional check:*  $T^{-1}$  times  $L^2$  equals  $L^2 T^{-1}$ .

### E5: Unit-circulation Biot-Savart reconstruction outside and on regularized filament fields

$$\mathbf{u}(\mathbf{x}) = \frac{\Gamma}{4\pi} \oint_C \frac{d\boldsymbol{\ell}' \times (\mathbf{x} - \mathbf{x}')}{(|\mathbf{x} - \mathbf{x}'|^2 + a^2)^{3/2}} \quad (15)$$

*Dimensional check:*  $\Gamma L$  times  $L$  divided by  $L^3$  gives  $L T^{-1}$ .

### E6: Finite-core line energy coefficient

$$E = \rho \Gamma^2 \ell_E, \quad \ell_E = \frac{1}{8\pi} \oint \oint \frac{d\boldsymbol{\ell} \cdot d\boldsymbol{\ell}'}{\sqrt{|\mathbf{x} - \mathbf{x}'|^2 + a^2}} \quad (16)$$

*Dimensional check:*  $\ell_E$  has  $L$ ;  $\rho \Gamma^2 \ell_E$  gives  $(M L^{-3})(L^4 T^{-2})L = M L^2 T^{-2}$ .

**E7: Relative-equilibrium decomposition**

$$\mathbf{u}_{\text{ind}}(\mathbf{X}) = \mathbf{U} + \boldsymbol{\Omega} \times (\mathbf{X} - \mathbf{X}_0) + \lambda(s)\mathbf{t}(s) + \mathbf{r}_{\perp}(s) \quad (17)$$

*Dimensional check:* each term has velocity dimensions  $L T^{-1}$ ;  $\lambda$  has  $L T^{-1}$ .

**E8: Exterior circulation period and linking consistency**

$$\frac{1}{\Gamma} \oint_{\gamma} \mathbf{u} \cdot d\boldsymbol{\ell} \approx \text{Lk}(\gamma, C) \in \mathbb{Z} \quad (18)$$

*Dimensional check:* line integral has  $L^2 T^{-1}$  and division by  $\Gamma$  is dimensionless.

**E9: Local exterior potential-flow audit**

$$\nabla \cdot \mathbf{u} = 0, \quad \nabla \times \mathbf{u} = \mathbf{0} \quad \text{away from vortex support} \quad (19)$$

*Dimensional check:* both divergence and curl of velocity have  $T^{-1}$ .

**E10: Exterior kinetic-energy fraction proxy**

$$f_{\text{ext}} = \frac{\int_{V \setminus V_{\text{core}}} \frac{1}{2} |\mathbf{u}|^2 dV}{\int_V \frac{1}{2} |\mathbf{u}|^2 dV} \quad (20)$$

*Dimensional check:* identical energy-density-volume factors cancel, leaving a dimensionless fraction.

**E11: Far-field localized-source exponent estimator**

$$\log u_{\text{rms}}(r) = b - p \log r, \quad p = -\frac{d \log u_{\text{rms}}}{d \log r} \quad (21)$$

*Dimensional check:* the slope of logarithms of consistently scaled quantities is dimensionless.

**E12: Post-reveal Bernoulli exponent map for a steady irrotational exterior**

$$|\Delta p| \propto r^{-2p}, \quad |\nabla p| \propto r^{-(2p+1)} \quad (22)$$

*Dimensional check:* This equation maps only exponents; dimensional pressure normalization is not inferred by the blind fit.

**E13: PKLSA cross-source/provider classification agreement for one topology and one preregistered gate**

$$A_{T,g} = \frac{\max_s n_s(T, g)}{N_{\text{provider}}(T, g)}, \quad s \in \{\text{PASS, FAIL, INELIGIBLE, INTERNAL\_INCONSISTENCY}\} \quad (23)$$

*Dimensional check:* counts/counts is dimensionless

## Registered code-to-mathematics steps

1. **S1** [G0; equations E1]: Verify frozen protocol and resolve the exact source directory from `source_contract.json` plus `SST_WORKBENCH_ROOT`.
2. **S2** [G2; equations E5, E6]: Run self-contained Python FP64 circle sanity controls before external source scoring.
3. **S3** [P0; equations E2]: Evaluate the exact diagonal incompressible Cauchy map for zero/nonzero vorticity populations.
4. **S4** [P1; equations E3]: Evaluate  $\det(F)$  and the analytic material cross-section scaling.
5. **S5** [P2; equations E4]: Evaluate initial/final vorticity flux under the same exact stretch.
6. **S6** [H1; equations E5, E6, E7, E8]: Resample each closed source geometry, estimate finite-core thickness, and calculate original A016 H0-H4 observables at preregistered resolutions.
7. **S7** [P3; equations E5, E9]: Sample an exterior sphere and use centered finite differences of singular-kernel Biot-Savart velocity to estimate divergence and curl.
8. **S8** [P4; equations E8, E9]: Reuse meridian circulation periods together with nonzero exterior speed to test locally irrotational but globally nontrivial exterior flow.
9. **S9** [P5; equations E5, E10]: Integrate the finite-core velocity-energy proxy with deterministic Halton-ball samples at two sample counts and partition by the preregistered near-core mask.
10. **S10** [P6; equations E11, E12]: Fit log RMS speed versus log radius on preregistered far-field shells for nonzero-vector-area closed sources.
11. **S11** [B0; equations E5]: Build the instance-local C++17/OpenMP kernel and compare its FP64 Biot-Savart output against the Python FP64 reference with no fallback authority.
12. **S12** [X0; equations E13]: Group blind samples by opaque topology and upstream provider group, collapse same-provider carrier variants to one vote, exclude PKLSA-generated carriers from closure, and evaluate preregistered H2/H3/P3/P4/P5/P6 state agreement.

## 7 Exact algorithm: equation-to-code mapping

- S1. Verify the frozen protocol and resolve the pinned E010/PKLSA v0.4.0 version root from `source_contract.json`, `SST_WORKBENCH_ROOT`.
- S2. Execute self-contained Python FP64 circle controls.
- S3. Execute the exact diagonal incompressible affine Cauchy-map control for P0–P2.
- S4. Parse each centerline, remove a duplicated terminal point, and arclength-resample to the registered resolutions.
- S5. Compute the unchanged H0–H4 geometry, convergence, holonomy, relative-equilibrium, and symmetry observables.
- S6. Sample an exterior sphere and estimate divergence/curl by centered finite differences of the singular-kernel exterior field.
- S7. Estimate exterior kinetic-energy fraction with deterministic Halton-ball samples at two counts.
- S8. Fit far-field RMS speed on fixed radius factors for vector-area-eligible geometries.

- S9. Group opaque samples by topology and upstream PKLSA provider group and evaluate X0 across H2/H3/P3/P4/P5/P6; generated carriers are excluded from closure.
- S10. Build the instance-local C++17/OpenMP FP64 kernel and compare it with the Python FP64 reference.
- S11. Freeze all blind results before the framework permits reveal.

The gate graph is a fan-out graph after infrastructure prerequisites; it is deliberately not a serial scientific pass chain.

## 8 Source, provenance, and independence contract

The admissible external geometry layer is E010/PKLSA v0.4.0. A016 requires schema `PKLSA-HIGH-RES-QUALIFICATION` atlas version 0.4.0, one unique publication-ready release, and `qualification_gate_pass=true` for an admitted topology. KnotPlot/RidgeRunner, Fremlin, Gilbert, PTSA, and other routes may therefore enter through PKLSA rather than through a hard-coded single library. One PKLSA carrier is a sampling unit, but X0 uses the upstream `provider_group` as its vote unit. Multiple carrier variants from one provider are correlated and collapse to one vote. PKLSA-generated/derived carriers remain diagnostic and cannot close X0. Public outputs expose only HMAC-opaque topology/provider/source identifiers; the identity map is reveal-only.

## 9 Gate DAG and decision semantics

The infrastructure spine is  $G0 \rightarrow G1$  and  $G0 \rightarrow G2$ . H0–H4, P3–P6, and X0 require G0/G1/G2 but do not form a serial pass chain; X0 consumes the completed per-carrier classifications rather than requiring those classifications to PASS. P0–P2 and B0 require G0/G2. Thus H3=FAIL does not prevent P3–P6, X0, or B0 from running.

| Gate | Question                   | PASS                                                                       | FAIL/other                                      |
|------|----------------------------|----------------------------------------------------------------------------|-------------------------------------------------|
| G0   | frozen protocol integrity  | verified                                                                   | process stop before pipeline                    |
| G1   | PKLSA source admissibility | unique publication-ready v0.4.0 release with resolvable qualified carriers | source FAIL; source-dependent gates are not run |
| G2   | Python reference controls  | finite, positive energy; reversal error $\leq 10^{-12}$                    | numerical control FAIL                          |
| H0   | closed geometry/thickness  | $N \geq 32$ , closure ratio $\leq 2.5$ , thickness/edge $\geq 1.25$        | any eligible input fails                        |
| H1   | convergence                | energy and RE drift within 0.08                                            | any eligible input fails                        |
| H2   | holonomy/linking           | at least two valid loops and max error $\leq 0.08$                         | any eligible input fails                        |
| H3   | relative equilibrium       | normal NRMSE $\leq 0.18$                                                   | any eligible input fails                        |
| H4   | covariance                 | both errors $\leq 10^{-9}$                                                 | any eligible input fails                        |

|    |                                |                                                                             |                                                                                 |
|----|--------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| P0 | vorticity population           | Cauchy error $\leq 10^{-12}$ and no class switch                            | exact control fails                                                             |
| P1 | material volume                | $ J - 1  \leq 10^{-12}$                                                     | exact control fails                                                             |
| P2 | vorticity flux                 | relative drift $\leq 10^{-12}$                                              | exact control fails                                                             |
| P3 | exterior potentiality          | valid points $\geq 8$ ; div/curl residuals $\leq 0.08$                      | any eligible input fails                                                        |
| P4 | exterior circulation           | H2-valid period plus nonzero exterior speed                                 | any eligible input fails                                                        |
| P5 | exterior energy                | $f_{\text{ext}} \geq 0.01$ ; refinement drift $\leq 0.12$                   | any eligible input fails                                                        |
| P6 | far-field exponent             | $R^2 \geq 0.97$ , $2.4 \leq p \leq 3.6$ on eligible inputs                  | any eligible input fails; none eligible is unresolved                           |
| X0 | PKLSA cross-source consistency | at least 2 upstream providers and $A_{T,g} \geq 0.80$ for H2/H3/P3/P4/P5/P6 | provider dependence, internal provider inconsistency, or insufficient providers |
| B0 | C++ certification              | relative L2 $\leq 10^{-10}$                                                 | build/parity certification FAIL                                                 |
| G9 | reveal commitment              | framework verification                                                      | reveal denied if policy fails                                                   |

## 10 Numerical certification, convergence, and error budget

BASIC uses up to eight PKLSA carriers selected by topology/provider round-robin and  $N = 96, 160$ ; FULL uses all qualified carriers with  $N = 160, 256$ ; CERTIFY uses all qualified carriers with  $N = 256, 384$ . H1 compares finite-core energy and relative-equilibrium residual between the two registered resolutions. Holonomy uses at least 384 source points. P5 uses 1024 and 4096 deterministic Halton-ball samples. P3 uses an exterior centered-difference step  $h = 0.002R$ . P6 uses radius factors 4, 6, 8, 12 times the source bounding radius.

| Error source                     | Estimator                       | Acceptance                                    |
|----------------------------------|---------------------------------|-----------------------------------------------|
| Centerline discretization        | H1 relative differences         | $\leq 0.08$ for registered energy/RE measures |
| Holonomy quadrature              | integer absolute error          | $\leq 0.08$                                   |
| Exterior finite difference       | dimensionless div/curl residual | $\leq 0.08$                                   |
| Energy sampling                  | low/high $f_{\text{ext}}$ drift | $\leq 0.12$                                   |
| Far-field fit                    | linear-log $R^2$                | $\geq 0.97$                                   |
| Backend round-off/implementation | C++ vs Python relative L2       | $\leq 10^{-10}$                               |

## 11 Backend authority and parity

Python/NumPy FP64 is REFERENCE authority. The instance-local C++17 kernel is built through Framework v1.0.6 and receives CERTIFICATION authority only when the actual backend is serial



or OpenMP FP64 and parity passes. A Python fallback cannot close B0. GPU screening is not enabled in this A016 version.

## 12 Blindness, opaque identifiers, and reveal

Topology, carrier, source-family, provider-group, provenance-family, and independence identities are replaced in public sample metrics by 16-hex HMAC-SHA256 group identifiers derived from a private key. Raw PKLSA paths and geometry/source hashes remain in the private runtime identity map, which is excluded from the blind ZIP. Target-bearing assignment terms listed in the private forbidden-term policy are commitment-protected. Reveal is permitted after FULL or CERTIFY when G2 passed; unresolved scientific branches do not block reveal because the design intentionally preserves informative later measurements after earlier scientific failures.

## 13 Statistical/null model and multiplicity

This is primarily a deterministic numerical falsifier, not a parameter-fitting study. The per-carrier sampling unit is one PKLSA-qualified carrier. For X0, the vote unit is one upstream provider group within one topology; multiple variants from the same provider are not independent. H\*/P\* aggregate status is conservative: any eligible per-input failure makes the aggregate branch FAIL. P6 eligibility is fixed by vector-area ratio before exponent acceptance. P5 sampling uses deterministic low-discrepancy points rather than stochastic confidence intervals. No multiplicity correction is used because there is no null-hypothesis p-value family; all per-input values and thresholds are reported directly.

## 14 Results and automatic run provenance

### Automatic run provenance

| Field                   | Recorded value                                                                  |
|-------------------------|---------------------------------------------------------------------------------|
| Protocol bundle SHA-256 | 6f5063c5091300095e3513b62fa1e0dd96940451538404399524fb6a52866a9f                |
| Framework schema        | SST-FALSIFIER-PROTOCOL-2                                                        |
| Python                  | 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] |
| Platform                | Windows-11-10.0.26300-SP0                                                       |
| Output manifest SHA-256 | 342b72958d03df29bc05f1d63f9928aedf4223b53f37ea473d12c556d70b7677                |

### Gate ledger

| Gate | Status               | Question                                                                                      |
|------|----------------------|-----------------------------------------------------------------------------------------------|
| G0   | PASS                 | Is the frozen v1.0.6 protocol intact at runtime?                                              |
| G1   | FAIL                 | Is the publication-ready PKLSA v0.4.0 qualified geometry layer present?                       |
| G2   | PASS                 | Do analytic and numerical controls validate the Python FP64 reference?                        |
| P0   | PASS                 | Does the Cauchy vorticity map preserve the zero/nonzero vorticity positions?                  |
| P1   | PASS                 | Does the same material map conserve material volume to numerical tolerance?                   |
| P2   | PASS                 | Does vortex stretching preserve vorticity flux through the corresponding surfaces?            |
| H0   | NOT_RUN_PREREQUISITE | Do input centerlines satisfy closed-filament geometry and finite-thickness constraints?       |
| H1   | NOT_RUN_PREREQUISITE | Are finite-core energy and relative-equilibrium observables resolution-independent?           |
| H2   | NOT_RUN_PREREQUISITE | Do exterior meridian circulation periods agree with integer linking structure?                |
| H3   | NOT_RUN_PREREQUISITE | Are the relaxed centerlines relative equilibria modulo translation, rotation, and reflection? |

|    |                      |                                                                           |
|----|----------------------|---------------------------------------------------------------------------|
| H4 | NOT_RUN_PREREQUISITE | Do circulation reversal and mirror transformations satisfy the register   |
| P3 | NOT_RUN_PREREQUISITE | Is the velocity field outside the vortex support numerically divergence   |
| P4 | NOT_RUN_PREREQUISITE | Does the irrotational exterior retain nontrivial circulation periods link |
| P5 | NOT_RUN_PREREQUISITE | Is a resolved nonzero fraction of kinetic energy carried outside the pre  |
| P6 | NOT_RUN_PREREQUISITE | Does the closed-filament far field exhibit a stable localized-source deca |
| X0 | NOT_RUN_PREREQUISITE | For the same blind topology, do distinct upstream PKLSA provider g        |
| B0 | PASS                 | Does strict C++/OpenMP FP64 reproduce the Python FP64 Biot-Sa             |
| G9 | DEFERRED             | After the blind analysis, does commitment verification permit reveal a    |

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## Backend manifest

```
{
  "continue_after_scientific_fail": true,
  "cpp_certification": {
    "actual_backend": "openmp",
    "authority": "CERTIFICATION",
    "build_fingerprint_sha256": "220f952eead0f60cb84d08c60ce1a871f92a005d255f90169cbd3d2c2ca4e1",
    "device_name": null,
    "metrics": {
      "backend": "openmp",
      "openmp_compiled": true,
      "openmp_max_threads": 20,
      "precision": "float64"
    },
    "ok": true,
    "precision": "float64",
    "requested_backend": "cpp",
    "source_sha256": "ce1cc9f79dbae18db938657789d0ca3f70f26189c3b2b59c55a65796eed7865d"
  },
  "n_public_samples": 0,
  "n_sample_errors": 0,
  "python_reference": {
    "actual_backend": "python-numpy-fp64",
    "authority": "REFERENCE",
    "precision": "float64"
  },
  "schema": "A016-BACKEND-MANIFEST-3"
}
```

## Source manifest

```
{
  "manifest_sha256": "9b576d5937b79e9e0a1a674b6d8e15f22065c25c000ad4e2129dd0ec85c19ef4",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "E010",
      "evidence_class": "simulation",
      "exists": false,
      "independence_family": "pklsa-qualified-carrier-atlas",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\E_pipelines\\E010_pklsa_par",
      "provenance_family": "E010/PKLSA-v0.4.0",
      "registry_hint": {
```

```

        "catalog_id": "E010",
        "hits": []
    },
    "role": "qualified_geometry_provider",
    "source_id": "S0_PKLSA_V040",
    "version": "v0.4.0"
}
],
"workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}

```

The frozen text above is never rewritten to accommodate an outcome.

## 15 Interpretation, limits, falsifiable predictions, and minimal follow-up experiments

P0–P2 test the exact material-population logic of the smooth incompressible model, not a full time-integrated knot evolution. H3 remains the original strong static-centerline shape test and can fail while the matter–field partition branch remains informative. P3/P4 establish whether the computed exterior is locally potential yet globally circulation-bearing. P5 tests whether velocity-energy support extends beyond the near-core source zone. P6 supplies a frozen far-field exponent for later mechanism comparison. X0 separately determines whether the H2/H3/P3/P4/P5/P6 classifications are robust across distinct upstream PKLSA provider groups for the same topology; it is a representation-dependence test, not an assertion that topology uniquely fixes dynamics.

A minimal next experiment after v0.2.1 is a genuinely time-dependent Euler/finite-core evolution in which material tracers are initialized both inside and outside a vortex tube, then tested for support-class preservation, circulation conservation, and topology change under simultaneous spatial and temporal refinement. A second independent experiment should reconstruct pressure from the time-dependent velocity field rather than infer pressure solely from a steady exterior relation.

## 16 Reproducibility and exact commands

Place the instance under the A016 Workbench family and keep the frozen framework under 06\_templates/SST\_Fal. If needed, set SST\_WORKBENCH\_ROOT. Then run:

```

run_all.cmd FREEZE
run_all.cmd BASIC
run_all.cmd FULL
run_all.cmd CERTIFY
run_all.cmd REVEAL

```

The first three scientific modes write blind packages; reveal is commitment-verified and separately packaged.

## 17 References

### References

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- [2] A.-L. Cauchy (1815), “Théorie de la propagation des ondes à la surface d’un fluide pesant d’une profondeur indéfinie,” historical source for the Lagrangian vorticity invariants; modern formulation summarized in standard vortex-dynamics texts.
- [3] H. K. Moffatt (1969), “The degree of knottedness of tangled vortex lines,” *Journal of Fluid Mechanics* **35**, 117–129. DOI: <https://doi.org/10.1017/S0022112069000991>.
- [4] P. G. Saffman (1992), *Vortex Dynamics*, Cambridge University Press. DOI: <https://doi.org/10.1017/CB09780511624063>.
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